

Siddarth M

📍 Chennai ✉ siddarthmofficial@gmail.com ☎ 9894297072 📄 in/siddarth-m-43ba61246 🌐 siddarth-py.github.io/

SUMMARY

Experienced Software and Systems Engineer with 5+ years in system and software development. Full Stack Developer with strong backend expertise in Python and Django, and hands-on experience building responsive frontends. Proven ability to design RESTful APIs, optimize databases, and deliver scalable applications

EDUCATION

B.E-Computer Science

K.Ramakrishnan College Of Technology · Trichy,Tamilnadu · 2014-2018

SKILLS

Programming Languages: Python, C, Embedded C, JavaScript

Backend Frameworks: Django, FastAPI, Flask

Frontend Technologies: HTML, CSS, Bootstrap

AI / Computer Vision: PyTorch, Segment Anything Model (SAM), OpenCV

Databases: MySQL, PostgreSQL, SQLite

Tools & Platforms:Postman,Docker,Git, GitHub, JIRA, IBM DOORS, Enterprise Architect, PTC

EXPERIENCE

Python full Stack Developer.

Mileveen Technologies

January 2025 - Present, Chennai

- Designed and developed a RESTful Booking Management API using FastAPI.
- Implemented CRUD operations for bookings (create, retrieve, update, cancel).
- Followed clean architecture with separation of router, use cases, and domain models.
- Used Pydantic models for request/response validation.
- Implemented proper HTTP error handling with status codes.
- AI-based application that analyzes images of retail shelves using computer vision techniques.

Software Engineer-

Ielektron Technologies

July 2023 - December 2024, Chennai

- Python, Django / FastAPI, HTML, CSS, JavaScript, PostgreSQL.
- Developed a full stack web application with backend APIs and frontend UI.
- Implemented CRUD operations and RESTful endpoints.
- Designed and optimized relational database schemas.
- Integrated frontend with backend APIs.
- Ensured responsive design for mobile and desktop.

Software Engineer

Webberax

November 2018 - May 2022, Chennai

- Writing the Functional TEST CASES For all features and all possible combination.
- Responsible for Writing the System level (SYS2) and Software Requirements(SWE1). Requirements(SWE1) .
- To controlling seat with different types of motors Track Recliner and Height via LIN & CAN Bus.
- System level Requirements(SYS2) and Software level Requirements (SWE1) Features like Welcome Recall, Storing Seat in different positions.
- Drafted and refined system-level (SYS2) and software-level (SWE1) requirements to support features such as Welcome Recall and multi-position seat memory, facilitating seamless integration with vehicle electronic control modules over LIN and CAN protocols.

PROJECTS

Retail_Shelf_Analyzer

Mileveen Technologies

Tech Stack: Python, FastAPI, PyTorch, Segment Anything (SAM), OpenCV, React, PostgreSQL, REST API

- Retail Shelf Analyzer an AI based application that analyzes images of retail shelves using computer vision techniques.
- The system uses the **Segment Anything Model (SAM)** to automatically segment product items from shelf images.
- Based on the generated segmentation masks, the system identifies product items, estimates stock count, detects empty shelf space, and provides analytical insights through a visual dashboard.

FastAPI Booking Management API

Mileveen Technologies

Tech Stack: Python, FastAPI, MySQL, HTML, CSS, Bootstrap

- Designed and developed a RESTful Booking Management API using FastAPI.
- Implemented CRUD operations for bookings (create, retrieve, update, cancel).
- Followed clean architecture with separation of router, use cases, and domain models.
- Used Pydantic models for request/response validation.
- Implemented proper HTTP error handling with status codes.

Django Employee Management System

Ielektron Technologies

Tech Stack: Python, Django, MySQL, HTML, CSS, Bootstrap

- Developed a full-stack CRUD-based Employee Management System using Django, enabling efficient creation, update, and management of employee records.
- Implemented secure authentication and role-based access control to restrict functionality based on user roles.
- Designed relational database models and optimized queries using Django ORM to ensure efficient data retrieval and integrity.
- Built a responsive and user-friendly frontend using HTML, CSS, Bootstrap, and JavaScript.
- Ensured secure form handling with proper validations, CSRF protection, and server-side checks.

Django REST API with JWT Authentication

Ielektron Technologies

Tech Stack: Python, Django, Django REST Framework, JWT, SQL

- Built secure RESTful APIs using Django REST Framework with JWT-based authentication for access and refresh token management.
- Implemented complete authentication workflows including user registration, login, token refresh, and role-protected endpoints.
- Designed REST-compliant API structures with proper HTTP methods, status codes, and API versioning for long-term scalability.
- Tested and validated API endpoints using Postman, ensuring correct request handling, authentication flows, and error responses.
- Designed and implemented a scalable backend architecture with clear separation of routing, business logic, and data models to improve maintainability and extensibility.

Autopark, Park assist

Webberax

- Auto Park system park vehicle automatically using guidance technology and cameras.
- Park Assist helps the driver park the vehicle using guidance technology and cameras.
- Understanding system-level requirements and releasing base versions.
- Responsible for creating and maintaining the Software Requirement Specification (SRS) Document (SWE1).
- Collaborating with stakeholders to gather and understand requirements. • Reviewing each requirement for accuracy and feasibility.
- Addressing and resolving review comments to ensure clarity and completeness.
- Facilitating effective communication between different teams to align on project goals.
- Conducting regular meetings and follow-ups with stakeholders for updates and feedback.
- Coordinated cross-functional collaboration to define, refine, and document comprehensive system requirements for autonomous parking and driver assistance technologies, ensuring stakeholder alignment and successful project milestones.

TWIN BLDC MOTOR

- The BLDC motor drives the air compressor, which generates compressed air.
 - The electronic commutation of BLDC motors allows for precise control over the motor's speed and torque.
 - Initialized hardware resources, including clocks and peripherals, ensuring efficient system operation.
 - Ensured system integrity using features such as CRC checks and secure boot mechanisms.
 - Developed System Basis Chip (SBC) control on the RH850 microcontroller to monitor power supply and protect against overvoltage, undervoltage and overcurrent by cutting off power to the microcontroller.
 - Maintained the ASPICE model, leading to consistently high-quality software development and compliance with industry standards, verified by successful audits.
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